

W-08 Real-Time Poverty Monitoring Using GDP Nowcasts

Evidence from India

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The Case for Real-Time Poverty Monitoring

Two forces make a monthly poverty read valuable: the case for nowcasting itself, and India's unusual data gap.

QUESTION 1

Why Nowcast Poverty?

Survey lag

Household expenditure surveys are the gold standard but are costly, infrequent, and released with a multi-year lag.

Policy relevance

Poverty rates and welfare aggregates drive resource allocation toward the Sustainable Development Goals.

Growth–poverty link

Decades of evidence show a strong, stable correlation between GDP growth and poverty reduction.

GDP itself is slow

Even official GDP lags ~6 weeks plus revisions; many low-income countries publish it only annually.

QUESTION 2

Why India Specifically?

Global weight

India's population size gives its poverty trajectory outsized influence on regional and global aggregates.

An 11-year data gap

No official consumption survey existed between 2011–12 and the 2022–23 HCES.

A clean validation test

The 2022–23 HCES gives a rare real benchmark to backcast against and validate the framework.

A transferable template

A framework proven on India is designed to generalize to other data-constrained economies.

From High-Frequency Data to a Monthly Poverty Read

How a monthly GDP nowcast becomes a monthly poverty estimate, and the policy decisions it feeds.

Illustrative transmission chain:

1. **High-frequency indicators update monthly** →
2. **Seven models nowcast GDP, updated monthly** →
3. **GDP growth scales the welfare distribution** →
4. **A monthly updated poverty rate emerges**

Policy questions this raises

Social protection / fiscal policy

How should safety-net transfers and subsidy budgets be sized against a poverty shock's expected depth and duration?

SDG and development tracking

Are anti-poverty interventions on pace, without waiting years for the next survey to confirm or refute it?

The common constraint: every one of these decisions needs today's poverty read, but official household surveys arrive years apart. In India's case, an 11-year gap between rounds.

LITERATURE REVIEW

Two Disconnected Strands

Poverty nowcasting and GDP nowcasting have each matured on their own. This paper connects them.

GDP-Based Poverty Nowcasts

Mahler, Castañeda Aguilar & Newhouse (2022) · World Bank Poverty & Inequality Platform

Main contribution: distribution-neutral scaling of the last observed welfare distribution by GDP per-capita growth performs nearly as well as much richer machine-learning models.

Literature gap: relies on ex-post official GDP data or biannual WEO forecasts, not real-time nowcasts, so it cannot track within-year fluctuations.

Real GDP Nowcasting

Cashin et al. (2025) · Bridge / MIDAS / factor-model / machine-learning literature

Main contribution: high-frequency econometric and machine-learning techniques deliver monthly GDP updates well ahead of official quarterly releases.

Literature gap: the nowcasting output stops at GDP itself: it is rarely translated into real-time welfare or poverty estimates.

Where the gap is:

Despite robust literatures on each side, growth-based poverty nowcasting and real GDP nowcasting remain largely disconnected.

How this paper bridges the gap:

A three-layer framework feeding monthly GDP nowcasts directly into a distribution-scaling poverty model, plus a Big Data value-added test.

Contribution: to the authors' knowledge, the first framework to link real-time GDP nowcasts to poverty monitoring.

DATA SOURCES

Traditional Indicators

What goes into the model: 31 traditional Haver indicators across five categories

31 *monthly indicators pre-selected*

Sourced from Haver Analytics, spanning 2000 onward. LASSO regression selects the subset most informative for Indian GDP growth.

- Real Sector & Production Activity
- Prices & Financial Markets
- Monetary Policy & Money Supply
- External Sector & Trade
- Agriculture, Climate & Uncertainty

LASSO-selected indicators include:

Petrol, motor-spirit & LPG consumption



General industrial production index



CPI and stock price index



Broad money (NM3)



Real GDP of India's main trading partners (US, China)



DATA SOURCES

Non-traditional Data ("Big Data")

What goes into the model: three experimental Big Data sources.

1 Google Trends

45 keywords across 5 macro sectors (consumption, investment, government, exports, imports); PCA-summarized into one index per sector.

2 Nighttime Lights

Satellite-derived proxy for aggregate economic activity and urbanization, via Google Earth Engine.



3 Climate & Vegetation

Cumulative rainfall, average temperature, and the Enhanced Vegetation Index (EVI), weather shocks and agricultural activity.



Building the Three-Layer Framework

High-frequency indicators feed a three-layer pipeline from raw data to a monthly poverty rate.

1

Layer 1: Real GDP Nowcasting

LASSO variable selection, then seven nowcasting techniques (Bridge, PCA, 3PRF, DFM, Elastic Net, Random Forest, XGBoost) each produce a monthly-updated quarterly GDP nowcast.

2

Layer 2: Poverty Rate Nowcasting

Quarterly nowcasts are annualized into real GDP per-capita growth (net of population growth), a 70% pass-through rate is applied, and the last observed household welfare distribution is scaled forward.

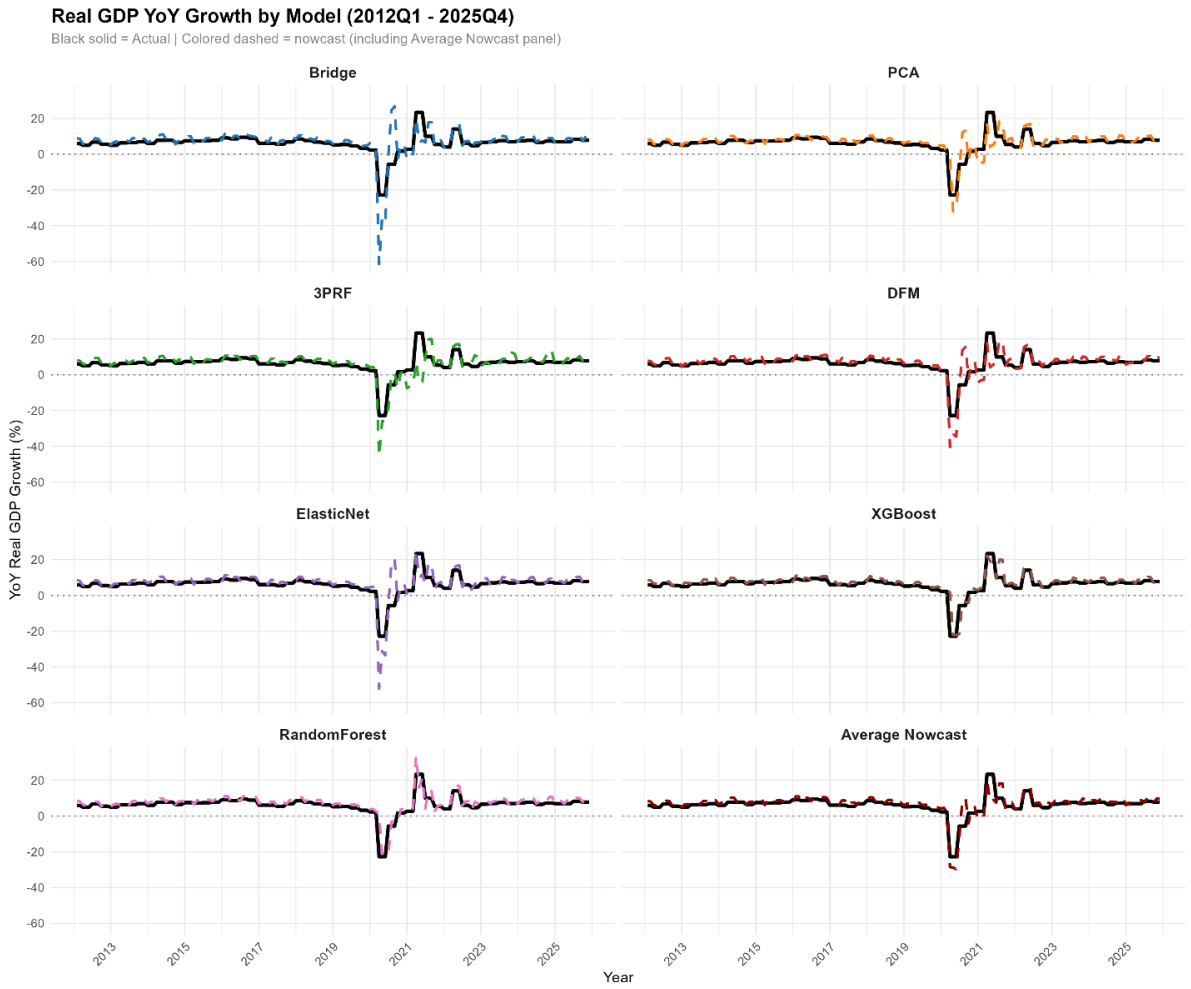
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Layer 3: Big Data Value-Added

Layers 1–2 are re-run with Google Trends and satellite indicators added, to test whether they improve nowcasting accuracy.

Layer 1: Real GDP Nowcast

How well the framework actually predicts GDP, out of sample.



GDP Nowcast Ranking

Out-of-sample, 2012Q1–2025Q4 (56 quarters / 168 monthly vintages)

✅ **Random Forest**	Lowest RMSE & Theil U2
XGBoost	Lowest MAD
Ensemble (7-model average)	Close to best ML, most stable
Elastic Net / PCA / DFM	Mid-pack
Bridge	Weakest on every criterion

OUT-OF-SAMPLE FORECAST PERFORMANCE

Layer 2: Poverty Nowcast

Out-of-sample poverty nowcast, by international poverty line, compared with the actual 2022–23 HCES

Poverty Line	$3.00/day	$4.20/day	$8.30/day
2022 Actual (HCES) %	**5.25**	**23.89**	**82.06**
Bridge	4.93	23.12	78.48
PCA	4.61	22.12	77.69
3PRF	4.69	22.38	77.90
DFM	4.60	22.08	77.66
Elastic Net	4.84	22.84	78.30
XGBoost	**4.95**	**23.19**	**78.54**
Random Forest	4.85	22.88	78.32
Average Nowcast	**4.78**	**22.66**	**78.13**

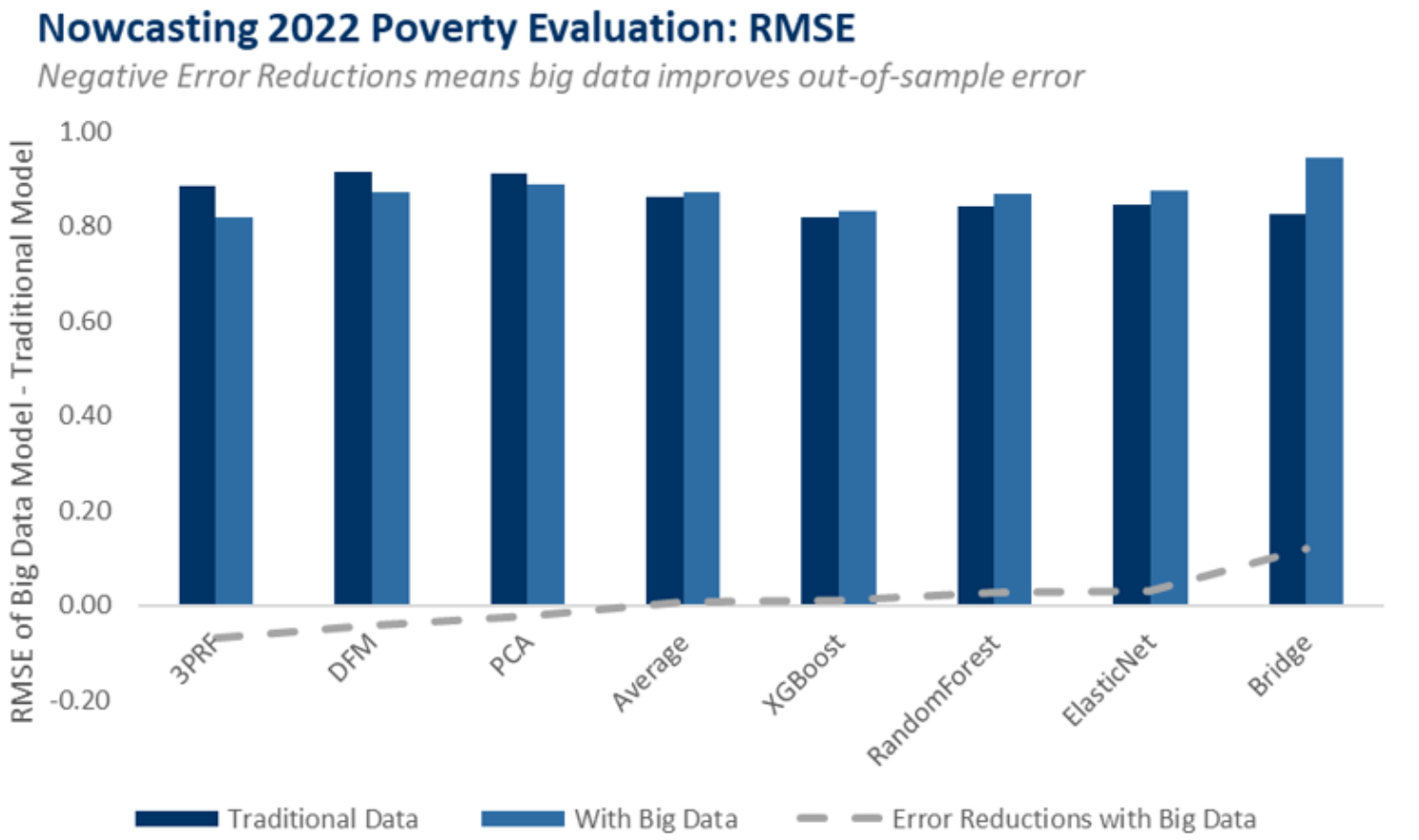
2022 Actual (HCES) Best-performing model (XGBoost) Average across all seven models

XGBoost is the most accurate model across all three poverty lines; the **Bridge model**'s nowcast is surprisingly close to actual despite weaker GDP-nowcast performance.

Source: Author's calculations, using 2022–23 HCES as the out-of-sample benchmark.

Layer 3: Value-Added of Big-Data

Testing whether Big Data improves the performance of poverty nowcast



How to read this chart

Traditional Data

Out-of-sample RMSE using only the 37+7 Haver indicators.

With Big Data

Same specification with Google Trends + satellite indicators added.

Error Reduction (dashed line)

RMSE(Big Data) − RMSE(Traditional); negative means Big Data lowers error.

3PRF, DFM and PCA see the largest error reductions from Big Data; **Bridge** shows the largest increase in error, with Average and XGBoost roughly unchanged.

CONCLUSION

A Monthly, Nowcast-Based Signal for India's Poverty Rate

Bringing the problem, the data, the model and the result together and what it means for policymakers.

The nowcasting story, in four steps

1 The problem

Official poverty surveys arrive years apart (India: an 11-year gap 2011–12 → 2022–23); existing GDP-based nowcasts rely on stale, ex-post GDP.

2 The data

31 monthly Haver indicators, LASSO-selected, optionally augmented with Google Trends and satellite imagery.

3 The model

A three-layer framework: 7 GDP nowcasting techniques feed a pass-through distribution-scaling poverty model, with an optional Big Data layer.

4 The result

ML models (Random Forest, XGBoost) deliver the best GDP and poverty accuracy

What this delivers, and where it goes next

Bottom line

Linking real-time GDP nowcasts to growth-distribution scaling delivers timely, operationally feasible poverty monitoring without waiting on the next survey round.

For policymakers

A between-survey read on poverty dynamics that supports SDG tracking, safety-net sizing, and real-time shock assessment.

Next steps

Extend to other data-constrained economies; test alternative pass-through/inequality assumptions; correct top-of-distribution bias.

Thank You!